Express-Kart: Comprehensive Project Documentation

1. Project Overview

Express-Kart is a multi-vendor e-commerce platform built with Flask and SQLite. It provides a complete ecosystem for buyers, sellers, and platform administrators.

2. Project Folder Structure

```
f:\FlaskApp\Ecommers
■   app.py                 # Main application entry point
■   config.py              # Application configuration (DB, Email, Razorpay, Admin)
■   auth_utils.py          # Authentication helper functions (OTP sending)
■   forgot_password.py     # Blueprint for password reset logic
■   init_db.py             # Database initialization script
■   ecommerce.db           # SQLite database file
■   requirements.txt       # Project dependencies
■■■■projects_document/      # Project documentation and guides
■   ■■■■README.md           # Quick start guide
■   ■■■■DOCUMENTATION.md    # Comprehensive technical documentation
■   ■■■■deployment_guide.md # PythonAnywhere deployment steps
■   ■■■■Ecommers_Project_Documentation.pdf # Exported PDF
■
■■■■static/                 # Static assets
■   ■■■■css/                # Stylesheets (style.css, auth.css, super_admin.css)
■   ■■■■uploads/            # User/Merchant uploaded images
■   ■■■■script.js           # Frontend interactive scripts
■
■■■■templates/              # Jinja2 HTML templates
■   ■■■■auth/               # Authentication templates
■   ■■■■merchant/           # Merchant panel templates
■   ■■■■super_admin/        # Super Admin panel templates
■   ■■■■user/               # User panel templates
■   ■■■■base.html           # Main layout template
■   ■■■■navbar.html         # Navbar component
■
■■■■utils/                  # Backend utility scripts
   ■■■■pdf_generator.py    # Invoice/Document PDF generation logic
```

3. System Architecture

The application follows a modular Flask pattern with the following components: - **Core Engine (`app.py`)**: Handles routing, session management, and core business logic. - **Database (`ecommerce.db`)**: Relational SQLite database. - **Frontend**: Jinja2 templates for dynamic HTML rendering, Vanilla CSS for styling. - **Authentication**: Bcrypt hashing and OTP-based verification for accounts.

3. Database Schema

- `users`: Stores customer information and hashed passwords.
- `admin`: Stores merchant and super admin accounts with status (active/pending/blocked).
- `products`: Product listings linked to merchants.
- `orders` & `order_items`: Transaction tracking.
- `cart`: Temporary storage for user shopping sessions.
- `wishlist`: User-saved products.

4. User Roles & Features

4.1. Customers (Users)

- Account registration with OTP.
- Product search and category filtering.
- Persistent shopping cart and wishlist.
- Secure checkout via Razorpay.
- Order history and invoice (PDF) downloads.

4.2. Merchants (Sellers)

- Self-registration (requires Super Admin approval).
- Dashboard with sales analytics and stock monitoring.
- Product CRUD operations (Create, Read, Update, Delete).
- Real-time order notifications.
- Profit margin tracking (default 20%).

4.3. Super Admin

- Full control over platform entities.
- Merchant approval/blocking system.
- Global product inventory management.
- Secure, hidden access route: `/admin-secure-access-xk9`.

5. Security Features

- **Hidden Admin Route**: The Super Admin login is not linked in UI to prevent brute force.
- **Brute Force Protection**: IP-based lockout after 5 failed attempts on the admin route.
- **Password Safety**: No passwords are stored in plain text.
- **OTP Verification**: Ensures valid email addresses for all registrations.

6. Development & Customization

Managing Super Admin Credentials

Super Admin credentials are set in `config.py`:

```
SUPER_ADMIN_EMAIL = "admin@123"
SUPER_ADMIN_PASSWORD = "admin123"
```

Update these values to change the administrative login.

Customizing Styles

The project uses two main stylesheets: - `static/style.css`: Main UI and responsive layout. - `static/auth.css`: Authentication specific styling.

7. Deployment Configuration

Refer to `deployment_guide.md` for detailed instructions on hosting the platform on PythonAnywhere.